Application Development Guide

Things to know and watch for when developing an L-PICK System application

Version 4.0

AIOI·SYSTEMS CO.,LTD.

Revision History

Date Printed	Version	Details
10/14/2005	Ver. 1.0	First Edition
11/22/2006	Ver. 2.0	Address change due to relocation
05/26/2008	Ver. 3.0	“WinCkX” changed to “Flowlight”
11/29/2013	Ver. 4.0	Changed “AiPlug” to “Lightstep”, “Echo” to “Reflexer”. Revised the description on how to calculate checking time of “Start Checking”. Added: “3.4 How to use Operation Instruction Command” Added: “5.3.6 How Operation Instruction affects performance”

■Contents

1. Overview 1
1.1 Basic Configuration 2
1.2 Devices 2
1.3 Communication Overview 3

2. Interface with the Controller 4
2.1 Lightstep 5
2.2 IoLibX 5

3. Commands 6
3.1 Command Format 7
3.2 Command Types 7
3.3 Process of Sending Commands 9
3.4 How to use Operation Instruction Commands 10

4. Development Procedure 12
4.1 Preparation for Development 13
4.2 Sending Commands from Flowlight 13
4.3 Design, Implementation, and Debugging of the Control Application 15

5. What to Watch for During the Development Stage 16
5.1 Minimum Essential Functions to Implement 17
5.1.1 Checking at the Start 17
5.1.2 Overcoming Obstacles 18
5.2 Implementation Pitfalls 20
5.2.1 Exclusion of Light Module Response Processing 20
5.2.2 Command Mismatch 20
5.3 How to Get High Performance 21
5.3.1 Minimize the frequency of communication between the control application and controller 21
5.3.2 Caution When Using the Block Number 22
5.3.3 Improved Speed When Using 2 Channels 22
5.3.4 Device Display Sequence 22
5.3.5 Using the Barcode Reader Interface 23
5.3.6 How to Send Operation Instruction to Get Better Performance 23
5.4 Caution on Commands 24
5.4.1 Values that Cannot Be Used 24
5.4.2 Values that Cannot be Used as String Type 24

6. Optional Functions of the Controllers and Light Modules 25
6.1 Fn Key Decrement Mode 26
6.2 How to Use the Light Modules' +/– Key 26
6.3 Automatic Control of the Signal Light 26
6.4 How to Use the Page-Turning Function 28

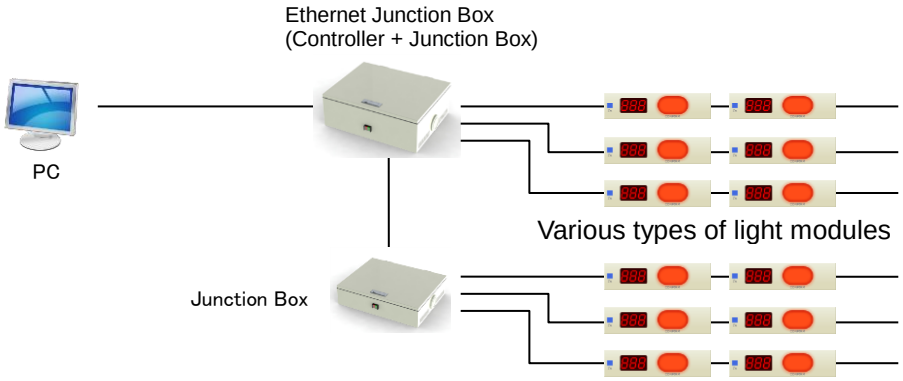
7. Debugging 30
7.1 Verify with Log 31
7.2 Number 0 Light Module 31
7.3 Operation Test Using Echo 31
7.4 Operation Test on Reflexer 31

8. Other 33
8.1 Other Manuals 34
8.2 Special Website for Users 34

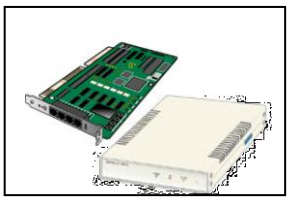
1. Overview

1.1 Basic Configuration

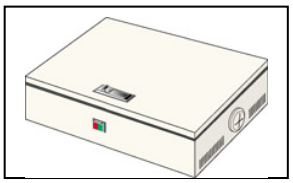
Although various devices used to construct a system will vary, the basic configuration is shown below. Each module connected to the junction box will have a 4-digit address from 0001 to 9999. Using the address, the application can easily control and give an operation instruction to various devices.



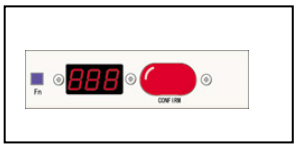
1.2 Devices



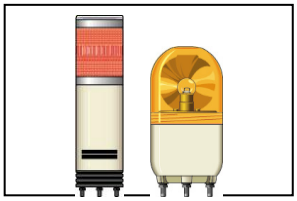
Controller
Serves as an intermediary between light modules and control application. After receiving an application command from the control PC, the controller sends a command to control light modules.
※Ethernet junction box has a built-in junction box .



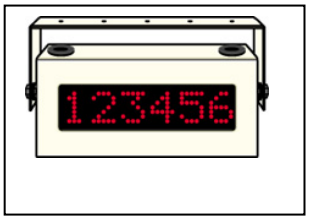
Junction Box
Provides power and sends communication signal to each light module.



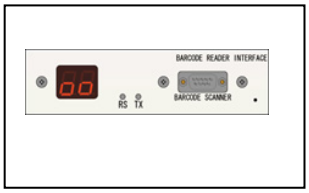
Light Module
Displays operation instruction to the worker. It has a CONFIRM button. Usable addresses are 0001 ~ 7999.



Signal Tower / Signal Light
Used to find out the condition of the system, such as if the system is in operation or whether there are errors. Usable addresses are 8000 ~ 8099.



Batch Display
A large display device, the batch display can display alphanumeric characters. It is used when displaying the content or progress of the operation. Useable addresses are 8100 ~8199. The usable addresses of the duct type batch display are 0001 ~ 7999.



Barcode Reader Interface
An interface that can connect to a barcode scanner, its usable addresses are 9101 ~ 9199.

1.3 Communication Overview

Control commands:

In order to perform functions such as turn the light module to ON or OFF, the control application sends commands in character string to the controller. The controller also sends commands to the control application on the information of whether the light module button has been pressed or not.

Detailed control functions of the controller:

The control application merely tells the controller which light module to light up or how it should light up through a command. According to the command, the controller controls the light module.

Capturing information after pressing the light module button:

When an operation instruction command is sent from the application side, the controller begins monitoring the specified light module's address according to the operation instruction command. When the button of the monitored light module is pressed, the controller checks it and reports by sending a command to the control application that the button has been pressed. After the button is pressed, the light module is no longer monitored. No matter how many times the button is pressed, nothing will happen until it receives the next operation command.

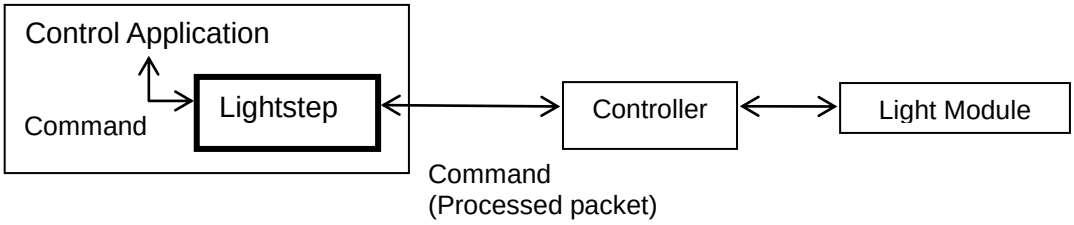
Inspection of a broken light module

If the light modules monitored by the controller are damaged (those that do not respond), the controller will send a command to report the light module address to the control application. If the light module is damaged, the control application can report the abnormality to the user.

2. Interface with the Controller

2.1 Lightstep

To send a command from the control application to the controller, the communication protocol and packet format must be specified according to the type of controller. It is also necessary to follow the specific procedures when sending a command. .NET Framework class library “Lightstep” combines these processes to allow users to easily send commands from the controller.*1



Lightstep monitors the connection/disconnection of the controller, the commands sent to the controller, and commands from the controller.
For details, please refer to the Programming Guide attached to Lightstep.

*1 Lightstep is available for purchase. The trial version is also available.

2.2 IoLibX

IoLibX is essential when using the PCI bus controller. IoLibX is a device driver configured with API (Iolibx.dll) so that it can be accessed from program.

When using Lightstep, IoLibX is not used directly from the control application. But when the PCI bus controller is used, IoLibX must be installed so that the controller is recognized by the operating system (OS).

3. Commands

3.1 Command Format

The command has the following form.

Header	Parameter

The header shows the type of command. The parameter differs depending on the command. The setup data is primarily entered in the parameter; however, there are commands with no parameter.

Example: Initialization command: **Z**
Block operation instruction command: P100000112345
(Header: P1, Parameter: 00000112345)

If the header begins with the upper case letter, the command is sent from the control application to the controller. If the header begins with a lower case letter, the command is sent from the controller to the control application.

In reality, when a command is sent to the controller, Lightstep creates a packet format according to the type of controller.

3.2 Command Types

The following is a general classification of commands. For further details, refer to the separate documents on the command reference.

Initialization command (Z)
The condition of each device is set to the same condition as when the power is supplied.

Operation Mode Setup Commands (M, Am1, Am2…)
Specify the light and buzzer operation (turn ON / OFF, flash, flash at a high speed, etc.)

Operation Instruction Commands (PP1, PP5…)
The controller begins monitoring if the button of the light module is pressed. And the lamp or buzzer is operated in the specified mode.

After the controller receives the **P1**, **P4**… commands from the PC, it turns on the light and begins monitoring the light module completion status (normal completion = **CONFIRM** button pressed, shortage = Fn key + CONFIRM button, and Abnormal).

Light Module Display Commands (PP2, P2)
Turn on the light module lamp or buzzer. The controller does not monitor whether the button of the light module has been pressed.

Termination Instruction Command (D)
The operation of all specified devices is terminated and lights are turned OFF.

Interface Command (G)
Opens and closes the barcode reader interface and RS-232C interface.

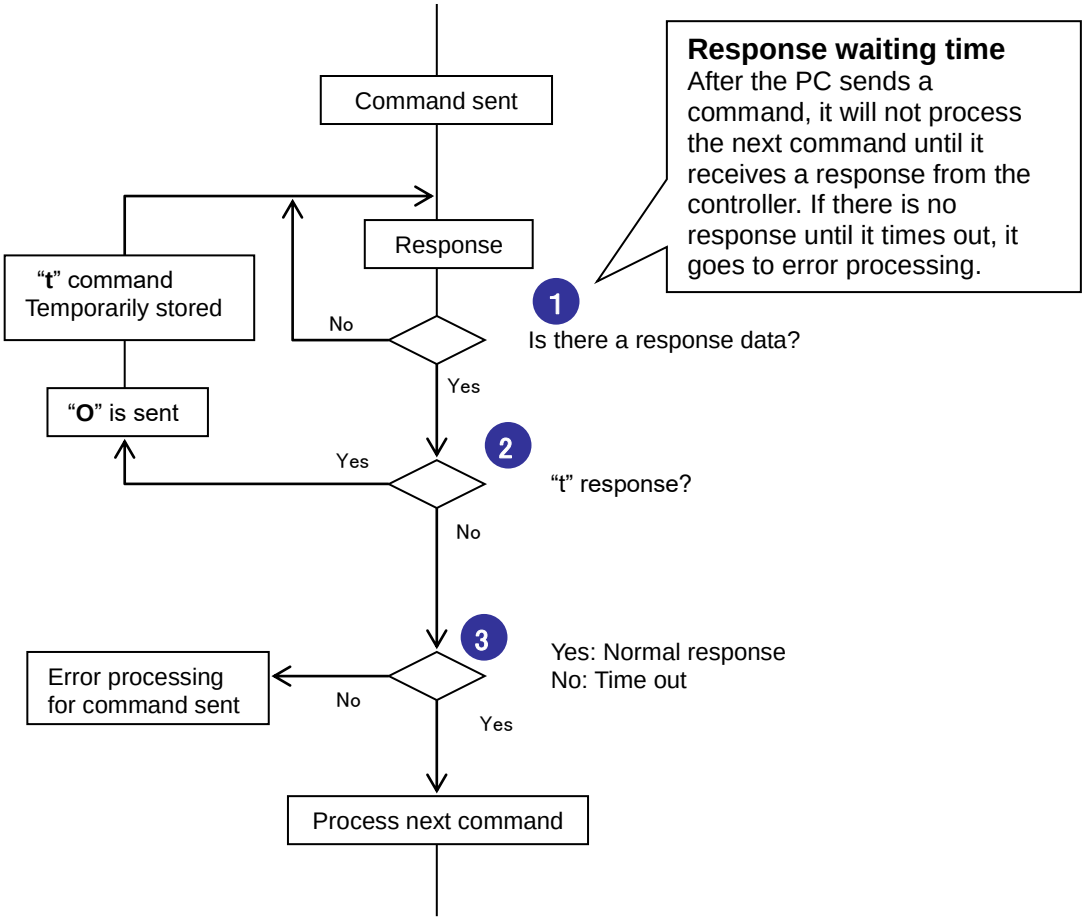
Maintenance Commands (A, Az, Ac)
Each module displays self address or address setup for each device.

Response Commands（o, n, ...）
A response command with respect to the sent command.

3.3 Process of Sending Commands

The sequence of a command sent from the PC for completion proceeds is shown below. After the PC sends a command, it always receives a response from the controller (see ① in the middle). The PC waits for a command from the controller (e.g., the "t" command). If the controller's response is "t", the PC sends the "O" command to the controller" (see ②). If the PC receives a response for a control application command, it analyzes the content (③) and the command that was sent is completed.

Lightstep performs this communication process internally. It determines and reports whether the sent command is successful or not. The "t" command is sent to the control application by way of "Event."



Note: If an "**n**" command is returned, the format or data range of the sent command may be incorrect. In order to easily find out that the "**n**" command was returned during debugging or after the software is released to the customer, create a way to find errors, e.g., by establishing a log.

3.4 How to use Operation Instruction Commands

Operation instruction commands are the most important commands in L-PICK system. These commands send operation instructions to the light modules and start monitoring when the “CONFIRM” button of the light module is pressed. This section explains basic information about these commands.

Sequence of Operation Instructions

When the control application sends an operation instruction command to the controller, the light module begins operating and the controller begins monitoring if “CONFIRM” button of the light module has been pressed.
When any action, such as pressing the “CONFIRM” button or pressing the “CONFIRM” button after pressing the “**Fn**” button, or when an abnormality happens on the light module, the controller terminates monitoring the light module and will tell the control application the monitoring is terminated. One cycle of an operation instruction consists of sending an operation instruction to the light module until receiving the completion status.

Recommended Operation Instruction Commands

AIOI recommends using the **PP1** command from among the several operation instruction commands that are provided. The **PP1** command is the command that can perform several settings required for operation instructions, including setting up the operation mode, display mode after pressing the “Fn” key and multiple addresses at the same time. This command can also be used to send an operation instruction to the light modules without specifying operation mode.

The **PP5** command is the command that can overwrite the **PP1** command (described later). The **P1** / **P4** / **P5** / **L** commands are old commands that are still compatible. There are some other operation commands including the ones that are particularly used for wireless light modules, specialized light modules, used only for page turning operation instruction, or specific to the type of module you use.

Cancellation of Operation Instruction

If you want to cancel the operation instruction that has already been sent to the light module, use the **D** (Termination instruction) command. The light module that has received the termination instruction will not be monitored and its lamp and buzzer will be turned off.

Duplication of Operation Instruction (Double sending)

You cannot send an operation instruction command to the light module that has already received an operation instruction command. You can only send an operation instruction command to the light modules that are on standby.
The standby status of a light module includes when the power is just turned on, the **Z** (initialization) command has been executed, the “CONFIRM” button has been pressed, or the termination command has been executed.

If you want to send another operation instruction to the light module which has received

one operation instruction, before its “CONFIRM” button is pressed; you can use the **PP5** command (rewritable operation instruction command). Another method is to send a termination instruction command to make the light module standby, and then send another operation instruction.

Sending an Operation Instruction to Light Modules Showing a Display

The **PP2 and P2** commands (light module display command) instructs the light module to only show or display. It does not monitor the light module.
To send an operation instruction to a light module in display-only mode, send the termination instruction command once, or use the rewritable command.

4. Development Procedure

4.1 Preparation for Development

Installation of Lightstep

If you use Lightstep, install Lightstep into the PC used for development.

Installation of IoLibX

If you use the PCI bus type controller, install IoLibX. After installing IoLibX, copy “IoLibX.dll”, which was copied to the folder for the installation, into the system directory or into the directory of the application to be developed. (IoLibX 3.0 or higher will be automatically copied).

Installation of Flowlight

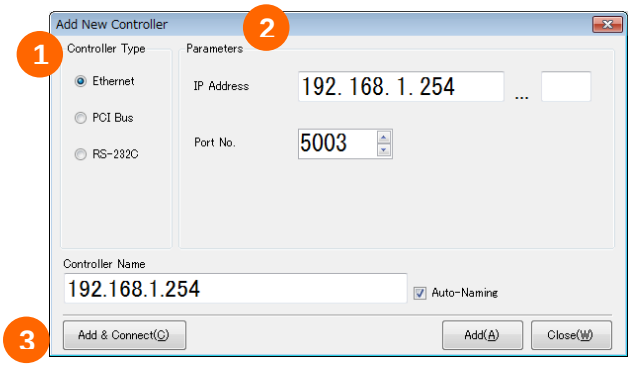
Flowlight is a light module-testing tool that enables a command to be entered and sent directly to the controller.

Debugging the Hardware Connection

Connect the controller to the PC used for development. Also connect the controller, junction box, light modules, and other devices. Please see the related user’s manuals and hardware manuals on how to connect to each device.

4.2 Sending Commands from Flowlight

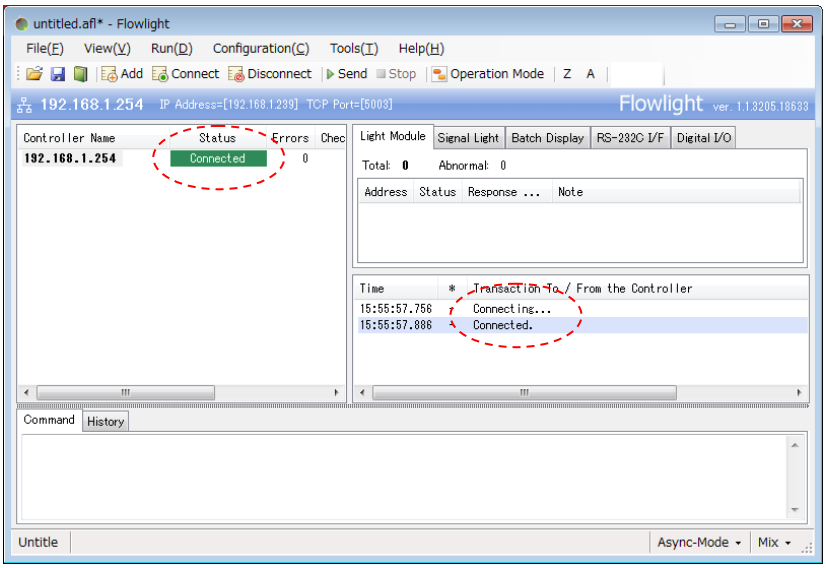
Before starting development, try to light up the light modules using Flowlight. This section explains how to use an Ethernet-type controller. The sequence of operation is similar to that of other types of controllers.



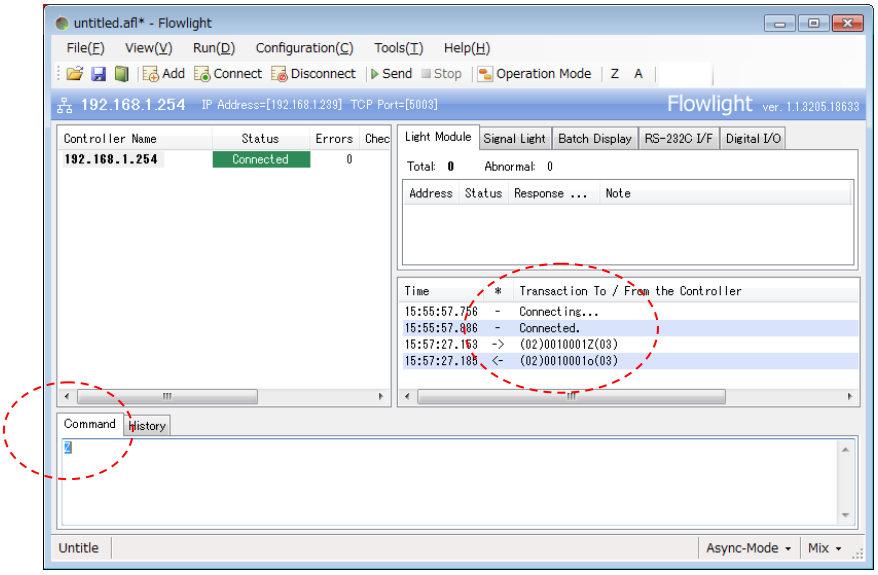
1. When Flowlight is started, the “Add New Controller” screen appears. Select a connected controller from among the controller types shown in the Controller Type column (①), then set up the parameters of the controller, which you will use, shown in column (②).

2. Click the “Add & Connect” button (③). The controller is added and connected to Flowlight.

3. When the connection is completed, the following screen should appear. If not, check to see if the parameters of the controller or physical connections are correct.



4. When you input a command in the Command section on the lower portion of the screen and press **Enter**, the command is sent to the controller. First try to send the **Z** command. The **Z** command is the initialization command of the device.



In the above screen, the communication log section shows **(02)0010001Z(03)** which indicates that the command has been sent to the controller. The next line shows **(02)0010001o(03)** which indicates that a response has been sent from the controller to Flowlight.

Next, when the **A** command is sent, all the connected light modules light up. When the CONFIRM button is pressed, the address will be displayed. The **A** command is the command for maintenance. When you want to turn light modules off, send the **Z** command.

Try sending various commands to find out the operations of the light module.

4.3 Design, Implementation, and Debugging of the Control Application

Get a good understanding of the ways to develop the system. Examine which commands to use to fulfill the requirements and start designing and implementing.

If Lightstep is not used when commands are sent to the controller, it is essential to develop an all-purpose function that processes a response from the module, such as light module, barcode interface reader, digital I/O, etc. Refer to Section 3 and the Command Reference.

If Lightstep is used, skip the development of the all-purpose function to respond and start developing the application control logic. Lightstep provides two sample programs: one is a simple program for controlling the light module and the other is a practical one for the control system. These sample programs can be used for development.

Because the hardware environment during the development stage cannot be set up in the same way as the actual installation site, only a limited number of light modules can be used for debugging. The installation will be facilitated if there is a smooth transfer from the development to the actual light module installation environment.

5. What to Watch for During the Development Stage

5.1 Minimum Essential Functions to Implement

5.1.1 Checking at the Start

If an external force is applied to a light module (e.g., hitting) the light module can break. Always check beforehand to make sure that each light module operates without problems. Below is a general guideline at the start of the operation.

(1) Send an operation instruction command to all the light modules that are installed. Make the batch display “888888” because the software cannot recognize an abnormality. A check is performed visually. If a barcode interface is used, open the interface and monitor the data input.

(2) Next, observe the light module response at a certain fixed interval. If there is abnormality, it is reported to the control application. The control application will display addresses and locations of the abnormal light modules on the screen to show which light modules are abnormal. It is practical to have a printer with printing capability to print out this information.

(3) While light modules are being checked and no error message is received, the module is considered to be normal.

Checking Time

The time required for checking depends on the number of installed light modules. Below is an example of calculating the checking time of module with no abnormal light modules:

Checking time (sec) = Number of light modules (*1) x 0.007 (sec)

(*1): Number of light modules controlled by one controller

The time required to check the status of one light module is about 0.007 sec.

If one light module is abnormal, it will be rechecked before judged to be abnormal. This takes about 0.5 sec.
If all the light modules are abnormal, the rechecking time is calculated as follows:

Checking time (sec) = Number of light modules x 0.5 (sec)

If the system configuration includes barcode interfaced, the controller checks 5 light modules at a time and then checks all the interfaces that are open.
If there is no abnormality the checking time is calculated as follows:

Checking time (sec) = Number of light modules x 0.007 (sec) + number of light modules ／ 5 x number of barcode interfaces x 0.007 (sec)

If a barcode reader interface is judged to be abnormal, further monitoring is not made, but this makes the calculation of checking time complicated. Below is the maximum required time based on the final retry of light modules and barcode interfaces when they are judged to be normal.

Checking time (sec) = Number of light modules x 0.5 (sec)
+ number of light modules ／ 5 x number of barcode interfaces x 0.5 (sec)

For example, if the system consists of 500 light modules and 10 barcode interfaces, approximately12 minutes will be required for checking.

All the devices can be checked for abnormality at once if set to maximum checking time, but usually there is almost no abnormality. For this reason, it is not practical to needlessly lengthen the checking time. It is better to set up the number of units that can be tested at once and assign the necessary time for checking. If malfunction is detected, fix it and resume testing the remainder.

The following is a practical formula for calculating the checking time:
Checking time (sec) = A+B

A: Number of light modules x 0.007 (sec) + number of light modules / 5 x number of barcode reader interfaces x 0.007 (sec)

B: Number of malfunction light modules x 0.5 (sec) + number of malfunction light modules / 5 x number of malfunction barcode reader interfaces x 0.5 (sec)

A indicates the checking time when all the devices are normal
B indicates the checking time when a specified number of the devices malfunction.

If setup is made as follows:

Number of light modules: 500
Number of barcode reader interfaces: 10
Number of malfunction light modules: 100
and
Number of malfunction barcode reader interfaces: 5

Then the checking time will be approximately 110 seconds.

5.1.2 Overcoming Obstacles

Many years of trouble-free application does not mean that a problem will not occur. Some consideration must be made ahead of time when developing an application to anticipate known problems so that when a problem occurs, it is solved as quickly as possible while it is still small.

Damaged Light Modules

If light modules are damaged during operation, they will not light up when a command is sent. When it is not possible to visually find out if the light module is damaged, the operation can get disrupted. The control application, however, will be able to return an abnormal light module status to indicate when and at which operation instruction the light module became abnormal.

Be sure to display the abnormality on the screen to let the user know. It is important to determine beforehand with the system manager what action to take should an abnormality occur. For example, should the control application automatically stop or continue.

No Response from the Controller

Communication with the controller is vital in controlling the light module; otherwise the operation cannot proceed. If there is a problem with routing or environment between the controller and the control PC, the controller will not send a response for a sent command. The control application must check successes and failures for every command that is sent. If a sent command fails, stop the control. It is necessary to get it to the point whereby the user can manage the application situation. It is also essential to strive to design the control application that enables an operation to be restarted just before the obstacle occurred.

System Going Down from Power Failure and Other Unexpected Errors

Power outage occurs when the control PC is not equipped with a universal power supply (UPS) or the control application suddenly stops because of a potential bug. However, the control application should be designed in such a way so that the operation can restart at or close to the point of stoppage. In this way, the user alone can attain a normal system recovery. It is important to separate the temporary data (memory) and permanent data (database, etc.) so that even when the operation stopped, data integrity is not lost.

Operation Log

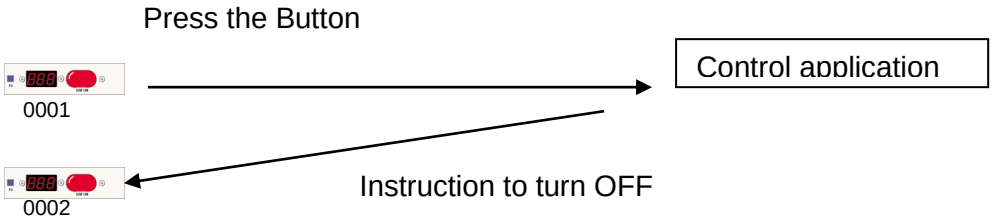
If there is abnormality, the cause of the abnormality must be pinpointed. Keeping a log will help determine the cause of the abnormality. Create a log without affecting the original processing performance and stressing the memory (hardware capacity). There is also a way to leave an option for leaving a detail log as it is being created.

By default, Lightstep keeps the log which can be used when needed.

5.2 Implementation Pitfalls

5.2.1 Exclusion of Light Module Response Processing

Let's say you flash several light modules and the system is set up in such a way that no matter which CONFIRM button is pressed, the processing is carried out in the same way. As an exclusion processing at this point, you may first think that it is better to press one of the buttons and not be able to press the remaining buttons by turning OFF the flashing lights.

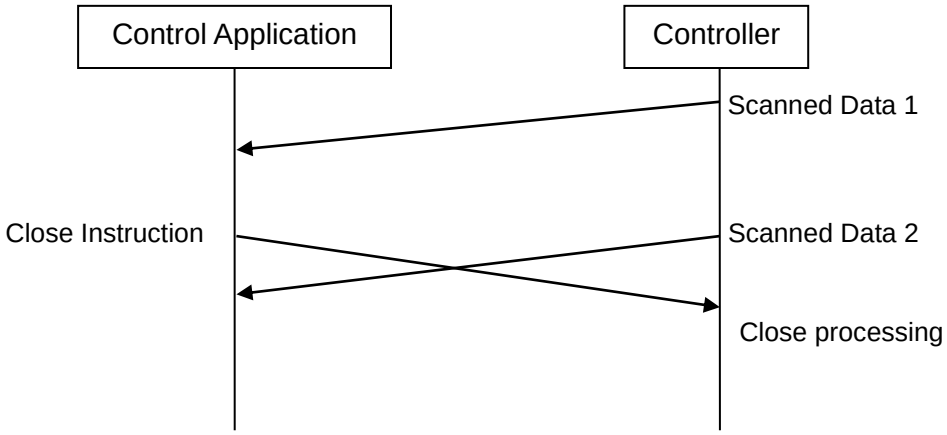


But, what happens when 2 light modules' buttons were accidentally pressed at the same time? The control application will get responses that the CONFIRM button was pressed from 2 light module addresses. When the control application responds to one CONFIRM button press and then tries to send an instruction to turn OFF the light of the other light module, it ends up sending the same command twice because the application has already received information from the first button press.
This should be avoided. Under certain circumstances, there may be problems, such as software stoppage or errors from losing data. Use "flags" to prevent the control application from performing duplicate processing.

5.2.2 Command Mismatch

Suppose you are using a barcode reader interface, and after receiving the data from the barcode scanner, you want to close the barcode interface to prevent further input from the scanner.

If the "G" command is used to close the barcode reader interface, the data cannot enter. However, it is important to consider the fact that from a timing mismatch, the barcode scanner data might come from the controller



The basic reason is similar to the exclusion mentioned above. Use flags and status as variables. Make sure that the control application does not receive unnecessary commands, which causes reckless processing, but that it processes appropriately by deleting or saving unnecessary data in the buffer, etc.

5.3 How to Get High Performance

5.3.1 Minimize the frequency of communication between the control application and controller

In order to execute one command, a command must be first sent to the controller and receive a response for the command, requiring two communications. A person may feel as if the command was sent instantly. However if there are several hundred commands, it may take 2 to 5 seconds to process, which is impractical.

Among various commands, there are some that can specify a command to several addresses* and send an operation instruction all at once. For example, instead of using the “L” command to send to the individual light module, it is better to use the “P1” command to send to all 100 light modules at the same time. One command takes the place of many.

* There is a limit on the number of addresses that one command can specify. For details, refer to the Command Reference.

Specifying the Operation Mode

Light module operation modes, such as turning the light ON or OFF, sounding or not sounding the buzzer, can be specified by the operation modes. How the operation mode commands are used can make a difference on the performance.

When the same operation mode is always used for the same light modules that are being controlled, it is a good idea to send the specified operation modes all at once. When all the operation modes are set up at once, the light modules will maintain the last operation mode setting until either the initialization command is executed or the power is turned OFF. It is not necessary to send an operation mode specification command

for each operation instruction.

If the controlled light modules must display various operation modes, using one “PP1” operation setup instruction command will set up the mode and specify the operation instruction.

Using the Wildcard

There are commands that can use the wildcard “?” for addresses. It is used when setting up the operation mode to all addresses but has limitations when used for different types of system devices.

5.3.2 Caution When Using the Block Number

With the block operation instruction command, the block number can be specified and used to send an operation instruction. The controller can automatically turn the signal light for that block number to ON or OFF. (For further details of this function, refer to Section 6.3, Automatic Control of the Signal Light).

In the case of an RS-232C or PCI bus controller, because light modules are monitored by each block, it may take time for the control application to receive information after the CONFIRM button has been pressed. For these controllers to control light modules, fix the block number to “00”. Because the automatic function to turn the signal light to turn ON and OFF cannot be used, have the control application send a command to turn the signal light ON and OFF. There is a big merit in having a function that turns the signal light ON and OFF automatically.

This problem does not come up when using the Ethernet controller.

5.3.3 Improved Speed When Using 2 Channels

Improved speed is obtained only when, through software, two channels are used simultaneously with the PCI bus controller.

If an operation instruction is sent to a Channel 1 light module address and Channel 2 light module address, the speed to process a command from the controller’s side will be much quicker if the operation instruction command for the Channel 1 and Channel 2 addresses is specified alternately rather than specified in order. (Refer to the Command Reference for more information.)

5.3.4 Device Display Sequence

In case a batch display is used to display the operation code or shop code for operation instruction, better performance is obtained by displaying the data on the batch display before making the light modules light up instead of displaying both the batch display and light modules at the same time.

First, the worker checks the code displayed on the batch display then looks for the light

modules that are lit. The worker turns ON the batch display and watches the batch display. During this time, the operation instruction is sent to every light module. The time it takes for the worker to check the batch display then turns to look at the light modules can be put to better use (for sending an operation instruction), even if there are many light modules.

Check the application method one more time. By looking at this from the worker's viewpoint may give you hints on how to improve the performance.

5.3.5 Using the Barcode Reader Interface

For systems using the barcode reader interface and RS-232C interface, the performance will improve if "open" display is reduced as much as possible. If the interface is open, the controller will monitor the data input rather than the light modules. Minimizing the monitoring of interfaces can speed up the transmission of CONFIRM button press information to the control application.

For systems where picking starts after scanning the barcode and the next barcode is not read until the picking is finished, close the barcode reader interface after scanning the barcode. Reopen only after picking is completed and the next barcode can be scanned.

5.3.6 How to Send Operation Instruction to Get Better Performance

It is not necessary to send an operation instruction to all devices. For optimal performance, send a command only to the light modules that are used. This applies also to the barcode reader interfaces and batch displays. The more devices that a controller controls, the more time is required for the application to get information.

Besides the operation instruction command, there are also commands that only make light modules to light up. While an operation instruction command makes light modules light up and monitors if the CONFIRM button has been pressed or not, the **PP2** and **P2** commands will only light up the light module without monitoring and will not affect the performance. This also applies to the batch displays.

For more information, see "Light Module Display Commands" of Command Reference.

If you only want to light up light modules without monitoring, use the **PP2** and **P2** commands in order to minimize the impact on the performance.

5.4 Caution on Commands

5.4.1 Values that Cannot Be Used

Depending on how commands are programmed, 0x02, 0x03, 0x0D may be included in the command. These are special values used to specify the beginning (STX) and the ending (ETX) of the command packet. If these are used in the middle of the command, they will not be correctly recognized.

When different types of controllers are used, do not include the following data in the commands.

PCI bus controller and Ethernet controller: 0x02, 0x03
RS-232C controller: 0x0D

Be careful when the “M” command and the “P4,” “P5” commands are primarily used because they may include special characters.

5.4.2 Values that Cannot be Used as String Type

There are commands which can use the values in the ASCII code range &h81~&h9F, &hE0~&hFC. It may not be possible to specify as a string type or data may be garbled. But Lightstep can send commands with 81h by using “$81” in the string. If Lightstep is not used, the byte type string is used for sending.

Be careful when the **M** command and the **P4**, **P5** commands are primarily used because they may include special characters.

6. Optional Functions of the Controllers and Light Modules

6.1 Fn Key Decrement Mode

After the decrement mode is set to “Use” with the operation mode setup command and an operation instruction is sent, the quantity shown on the light module is reduced by 1 each time the light module’s “Fn” key is pressed. If it shows “0” and the “Fn” key is pressed, it returns to the original value.

When a change in the quantity is made with the “Fn” key and the CONFIRM button is pressed, the control application will be able to capture the quantity change. The normal response data (“h” and “t” commands) will return only the address and status data. When the number is revised with the “Fn” key, the changed value will also be added.

Example: When the CONFIRM button is pressed after the value of the light module address 1001 is changed to 12 (Using the Ethernet controller), the 5-digit displayed data will be enclosed in < >.

t100100< 12>

When the CONFIRM button is pressed with the “Fn” key operation mode set to the “decrement mode” and the “Fn” key is not used, the displayed data will not be added to the response data. When the “Fn” key is used, check to see if “ < > “ is followed by the status. If it is, check to see if the quantity has been changed.

When the “Fn” key is used, the displayed data is added. Even when used only once and return to the original value, the displayed data will be added to the response data.

6.2 How to Use the Light Modules’ +/– Key

The light module with the +/– key operates even if the operation mode is not specified. The data capture on the software side after the +/– key is used to change the quantity is the same as that of the “Fn” key decrement mode.

Specifying the Digits that Need to be Changed

For light modules with the +/– key, you can specify the range of digits that need to be changed. For example, to change the value within the 2 digit range, set the display data as a 2-digit number as in ” 02”. If it is ” 002”, the range is from 0 to 999.

6.3 Automatic Control of the Signal Light

The block number is specified with the block operation instruction command. The controller sends an operation instruction command to the light module. At the same time, it automatically lights up the signal light whose last 2 digits of the address are equivalent to the block number.

Example 1:

When an operation instruction is sent to the light modules 0001 and 0002 with the block number 01, they will be activated. At the same time, the signal light 8001 will light up.

When the light modules that received the operation instruction with the same block number are terminated (by the CONFIRM button press or from abnormality), the signal lights with the same last 2 digits as the block number will automatically turn OFF.

Example 2:

In Example 1, when the CONFIRM button is pressed for address 0001, the signal light 8001 will stay lit. When the CONFIRM button of 0002 is then pressed, the signal light 8001 will automatically turn OFF.

Note: When signal lights are used with RS232C and the PCI bus controller, and when there are many block numbers, poor performance results (Refer to Section 5.3.2 on “Caution When Using the Block Number.”

6.4 How to Use the Page-Turning Function

The page-turning function enables a maximum of 10 operation instructions to be sent to one light module with 1 command, and it can also capture the total operation result. Because the controller controls several operation instructions, the time between pressing the CONFIRM button and sending the next operation instruction is shortened. Consider the first operation instruction as 1 page. After this page is completed, the second operation instruction shows up - - - this function is like turning one page to the next page.

To use the page-turning function, a special controller is used. Compared to the regular controller, a limitation is that the special controller with the page-turning function can control few light modules. (For details, see L-PICK Controller Command Reference).

Page Turning Operation Instruction Command

Special commands, (**Pm and P6** commands) are used for sending multi-page operation instructions. These commands set up the operation mode as soon as the operation instruction is sent. Also, these commands can send operation instructions to multiple light modules *1. It can specify an operation mode to each address but cannot set up a separate operation mode for each page.

*1 Be careful of the limitation of number of light modules.

Light Module Operation

The light modules receiving a multi-page operation instruction will first display the first page of the specified display data. When the CONFIRM button is pressed, the second page is displayed. Also, if the “Fn” key is pressed, followed by the CONFIRM button, the next page will be displayed. (In the case of the +/– key, pressing the “Fn” key after page 2 will return to the previous page.) When the CONFIRM button of the last page is pressed, the operation result will be sent to the application.

Get the Operation Result

Just as you normally get the operation result, get the completion status by using the "**t**" command for the Ethernet type and the "**H**" command for the PCI bus type. The status will either be normal ("00") or abnormal ("01"). When the "Fn" key is used, the status will differ depending on the data type.

(1) When the "Fn" key used as a decrement key or when the +/– key is used

When the quantity is changed using the "Fn" key or the +/– key, the completion recognition command, the data format obtained by the completion recognition command will be different. The status is followed by " < ", page number (0~9), hyphen, and revised displayed data will be added. Only the page with the data after the quantity was changed will be added.

Example: When the quantity on page 1 for Address 1001 has been changed to 12, and on page 3, the quantity has been changed to 5 with normal completion.

```
h100100<0-    12><2-    05>
```

(2) Normal "Fn" key operation ("Fn" key without decrement mode, light modules without the +/– key)

In the case of format (1), the display data will be set to " 0" and sent.

7. Debugging

7.1 Verify with Log

When light modules are not lightning up as expected, first check to see if the command was sent as it was intended or if there were any mistakes in the command.

Lightstep has a function that can store the commands sent and received as text format Log files. By looking at the Log, the cause can be determined by looking at which commands were sent and received, and in what order.

A log of commands is also kept in the Ethernet controller. Refer to the manual, “Ethernet Controller User’s Manual.”

7.2 Number 0 Light Module

When the controller recognizes a light module error, it will display the address on 0000 of the light module. During debug, if a light module that is set up with the 0000 address is connected, an error can be easily located. For more information on errors, refer to the “Controller User’s Manual.”

The number “0” light module can only display the most recent error. If there is a series of errors, only the very last error can be displayed. However, if the controller is the Ethernet controller, a log is kept in the controller’s memory, and the file can be downloaded using FTP (refer to the “Controller User’s Manual”).

7.3 Operation Test Using Echo

Echo is a light module simulation software. Echo will show the status of all possible address ranges when light modules, batch displays, signal lights, and barcode interfaces are connected. Echo can perform an operation test for the same number of light modules that will be actually installed, thereby helping to discover as many bugs as possible before the pre installation testing.

Echo is a Windows-based software. Because the processing speed differs depending on the PC environment, the response speed cannot be simulated.

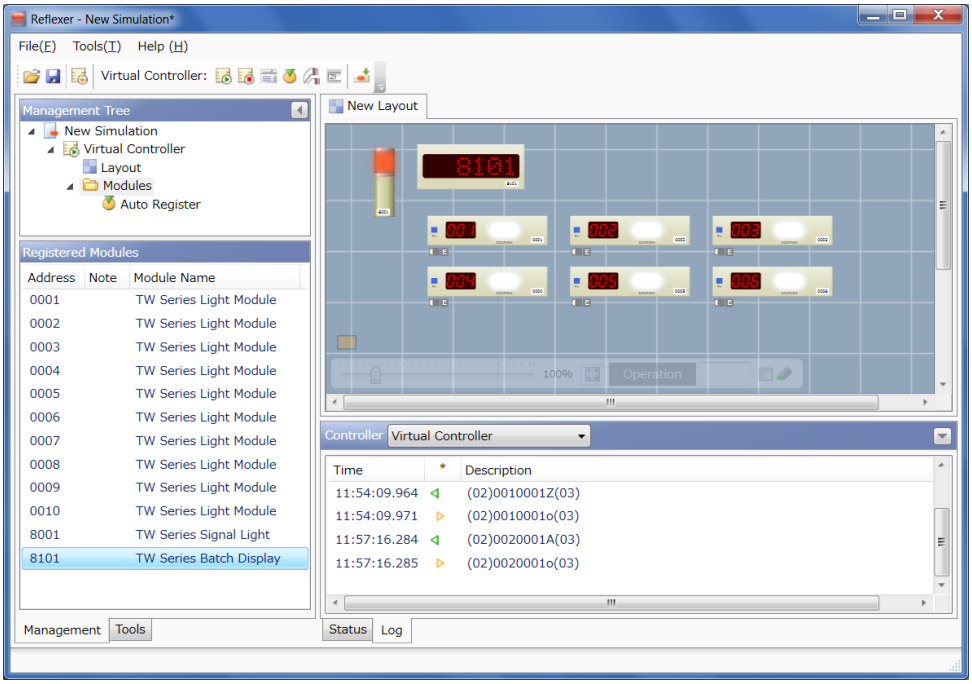
7.4 Operation Test on Reflexer

“Reflexer” is a simulation software program when the Ethernet controllers and standard light modules are used. It enables users to test the same number of the light modules and controllers that will be installed, thus helping to debug the system prior to operation.

Note:

Reflexer is software running on Microsoft Windows, and its processing speed varies depending on your PC environment. Therefore, response time will vary.

Reflexer is available for purchase. You can borrow Reflexer for one month free of charge. Please contact our sales person for more information.



Reflexer screen

8. Other

8.1 Other Manuals

In addition to this manual, other user manuals include:

“Controller User’s Manual”
Various types of controllers describing the initialization setup method and connection method.

“Command Reference”
Describes detailed explanation on how to use commands.

8.2 Special Website for Users

After registering with AIOI, users can download various files, including FAQ files. You can register with AIOI User’s Page at URL:

https://aioi-support.com/userpage_e/

[Note] The contents of this manual may be revised without prior notice.
No part of this manual may be copied or reproduced in whole or in part without the written permission of AIOI·SYSTEMS CO., LTD.

[Inquiries] Please send any inquiries regarding the contents of the specifications and manual of the TW Series, or application program development, by FAX or E-mail to AIOI·SYSTEMS CO., LTD. (info1@hello-aioi.com)

Application Development Guide

AIOI·SYSTEMS CO., LTD.
WiRA Omori Building, 8F
1-6-8 Omori Kita, Ota-ku, Tokyo, 143-0016 JAPAN
Tel. +81-3-3764-0028 Fax. +81-3-3764-7520
E-mail: info@hello-aioi.com

AIOI SYSTEMS USA CORP.
19925 Stevens Creek Blvd. Cupertino, CA 95014-2358
Tel. +408-973-7835 Fax. +408-973-7235
E-mail: info.us@hello-aioi.com

Copyright © AIOI·SYSTEMS CO., LTD.